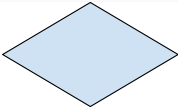


List Of Python Programs

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SR NO.	Questions / Programs
1	Print the simple Hello World Program.
2	Print User input Hello [Username] !
3	Type function.
4	Swap 2 nums without using 3rd num. [WITH 3-4 METHODS]
5	User-input 2 nums and swap. [WITH 3-4 METHODS]
6	Simple caculator.
7	User-Input : Make the calculator. [+ , - , * , /]
8	Simple Floor and Power
9	Fahrenheit to celsius program & celsius to Fahrenheit program [2 programs in 1]
10	Write a python Statement that calculate the area of a circle is $\text{pie} \times R^2$ where r is the radius of the circle and is 2, $\text{pie}=3.14$
11	Write a python statement that calculate the area of rectangle is where width is 4 and length is 7.
12	Write a python statement that calculates the circumference of a circle is $2 \times \text{pie} \times R$. where $\text{pie}=3.14$, $r=5$.
13	There are 5280 feet in a mile. write a python statement that calculate and prints the number of feet in 13 miles.
14	Write a python statement that calculate the future value of this money when compounded yearly at a rate of r percent intrest for : Compound Interest = $P(1 + r/100)^y - P$ P = Principal amount r = Rate of interest y = Time period in years
15	Write a python statement that calculate the perimeter of rectangle is $2w+2h$ where w and h are the length of its sides. Write a python statement that calculates and prints the length in inches of the perimeter of rentangle with sides of length 4 and 7 inches.
16	Write a python statement that calculate and print the number of seconds in 7hours 21minutes and 37seconds.
17	Simple Sep & End program.
18	Create Triangle Pattern using For loop.

19	WAP to print Armstrong Number using loop. [USER-INPUT]
20	WAP to print Fibonacci Series using loop. [USER-INPUT]
21	WAP to print Check number is +ve, -ve or zero. [USER-INPUT]
22	WAP to print Check number is ODD or EVEN. [USER-INPUT]
23	WAP to print Factorial of a number using for loop. [USER-INPUT]
24	i.) User input: Decimal to Binary, Octal, and Hexadecimal. ii.) User input: Binary, Octal, Hexa, 32 to Decimal.
25	Print first 10 natural numbers using while loop.
26	Calculate the sum of all numbers from 1 to a given number.
27	WAP to print multiplication table of a given number.
28	Display numbers from a list using loop.
29	Count the total number of digits in a number.
30	Print the following pattern: 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5
31	Print list in reverse order using a loop.
32	Display numbers from -10 to -1 using for loop.
33	Use else block to display a message 'Done' after successful execution of for loop.
34	Write a program to display all prime numbers within a range.
35	Reverse a given integer number
36	Use a loop to display elements from a given list present at odd index positions.
37	Calculate the cube of all numbers from 1 to a given number.
38	Find the sum of the series upto n terms LIKE : Series: 1 + 2 + 3 + 4 + ... + n
39	Menu-Driven program for Decimal to Binary, Octal and Hexa. [USING bin(), oct() and hex()]
40	Menu-Driven program for Binary, Octal, Hexa, 32 to Decimal. [USING int()]

41	Menu-Driven program for simple user input calculator. [+ , - , * , /]
42	Stack list. [PUSH, POP, DISPLAY]
43	Queue list. [PUSH, POP, DISPLAY]
44	Remove Duplicate values from a list using loop.
45	Merge list without duplicate elements in the merged list.
46	Program to count vowels and consonants in a string using two methods. [WITH MEMBERSHIP OP & WITHOUT MEMEBERSHIP OP.]
47	WAP to Check String is Palindrome or Not. [USER-INPUT]
48	WAP to Check number is Perfect number or not. [USER-INPUT]
49	Create Diamond pattern LIKE : 
50	Menu-Driven program : WAP to check entered number is : i.) a Palindrome or not. ii.) Armstrong or not. iii.) Reverse number iv.) Display number of digits without using len() v.) Sum of Digits.
51	Menu-Drive program : Tuple CURD [ADD, UPDATE, DELETE, DISPLAY]

Practice-1 :

- i.) How to find the number of elements in the list?
- ii.) How to find the largest and lowest value in the list?
- iii.) How to find whether the list is empty or not?
- iv.) How to find the first and last element of the list?
- v.) How to access an element of the list?
- vi.) How to modify an element of the list?
- vii.) How to concatenate two lists?
- viii.) How to add two lists element-wise in Python?
- 52 ix.) How to find the occurrences of an element in the Python list?
- x.) How to find the index of an element in the Python list?
- xi.) How to remove duplicate elements in the list?
- xii.) How to check if an item is in the list?
- xiii.) How to insert an item at a given position?
- xiv.) How to flatten a list in Python?
- xv.) How to convert a Python list to other data structures like set, tuple, dictionary?
- xvi.) How to apply a function to all items in the list?
- xvii.) How to filter the elements based on a function in Python list?
- xviii.) How Python lists are stored in memory?
- 53 WAP to merge the 2 dicts and create the new dict?
- 54 WAP to arrange values of dict in ASC order.
- 55 WAP to take User-Input & check if it is present in the dict or not
- 56 WAP to take User-Input n terms & print n: $n * 5$
- 57 i.) Add values into dict.
ii.) Get the keys.
iii.) Find min & max values.
iv.) Count the number of elements in dict.
- 58 WAP to take User-Input & multiply it with i.
- 59 WAP to Remove Duplicate values from dict.
- 60 WAP to Count for how many times the elements are repeated in dict.
- 61 WAP to take User-Input empid and if the salary is less than 25,000 then increment by 1,000 else no change.
- 62 WAP to create a student.txt and add few student's name with 3 subjects marks then read the file and check if student has less than 40 marks in any then print failed, else print pass.

63	Menu-Driven Dictionary CRUD program in Python [Create/Add, Read/Display, Update, Delete]
64	Create Employee Salary Slip using nested Dict.
65	Create Electricity Bill OR Adani Gas Bill using Dict.
66	Exception Handling: i.) Exception handling Structure. ii.) Built in exception and multiple except blocks. iii.) User defined exception [Create a valid named class & take user-input age and check eligible for vote or not.] iv.) Auto exception / Exception Aliasing [except Exception as e]
67	Function: i.) Normal Functions [create a function check a maximum num of 2 nums]. ii.) User-input Functions [create a function check a maximum num of 2 nums]. iii.) Parameterized Functions [for to check a maximum num of 2 nums]. iv.) Return Value Functions [for to check a maximum num of 2 nums]. v.) Default Parameters Function [3 args with default parameters to print your fullname(last_name + first_name)]. vi.) Advance Concept of Default Args [for 3 nums(x,y,z) after any default arg, all args must be default]. vii.) WAP to demonstrate nested function. viii.) WAP to demonstrate function as argument. ix.) WAP to demonstrate function can call another function
68	Assertion Exception Handling
69	Doc-String & Help Function for Documenting the Python Codes.
70	Recursion: i.) WAP to create a Fibonacci series using recursion. ii.) WAP to create a Factorial function using recursion. iii.) Sum of n natural numbers using recursion. iv.) Power calculation using recursion.
71	WAP to demonstrate call by value and call by reference.
72	Arbitrary Positional & Keyword Arguments [*argv & **kwargs]
73	WAP to perform Magic Method.
74	Modules: i.) WAP to print area and circumference of circle using module. ii.) WAP to print the random numbers using 'random' module. iii.) WAP to create a calculator using module. iv.) WAP to print string in lowercase using 'string' module.

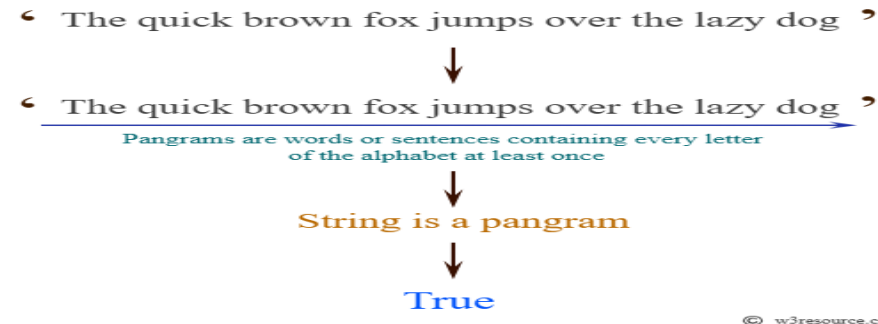
WAP to Print Pascal's Triangle & Also Write an Algorithm for it. LIKE :

75

```
      0C0
    1C0  1C1
  2C0  2C1  2C2
3C0  3C1  3C2  3C3
```

WAP to check user-input string is Pangram or not, and also Write an Algorithm for it. LIKE :

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77

WAP to generate random password that contains symbols, numbers, characters.

Practice-2 :

78

- i.) WAP function to find the maximum of three numbers.
- ii.) Sum all numbers in the list
- iii.) Multiply all numbers in a list.
- iv.) WAP to reverse a string.
- v.) WAP Function to check whether a number falls within a given range.
- vi.) WAP Function that takes a list and returns a new list with distinct elements from the first list.
- vii.) WAP to print the even numbers from a given list.
- viii.) WAP to Sort Hyphen-Separated Sequence of Words Alphabetically.
- ix.) WAP function to create and print a list where the values are the squares of numbers b/w 1 and 30 (both included).
- x.) WAP Function that takes a number as a parameter and checks whether the number is prime or not.
- xi.) WAP function to calculate the factorial of a number, (a non-negative int). The function accepts the number as an argument.
- xii.) WAP function that checks whether a passed string is a palindrome or not.

Files:

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- i.) File properties.
- ii.) WAP of file using readlines(), readline() and read().
- iii.) WAP of file using tell & seek.
- iv.) WAP of file for write, append, binary mode & comma separated values.

80	Menu-driven : File CURD operations [ADD, READ, UPDATE, DELETE, EXIT]
81	Menu-driven : File Binary CURD operations [ADD, READ, UPDATE, DELETE, EXIT]
82	WAP Scoping, Local Scoping, Global Scoping
83	WAP of file writelines() with File Exception Handling.
84	WAP of file first n line, last n line, Line count, character count.
85	Practice-3 : i.) Copy content from one file to another. ii.) Read a file line by line and store it in the list iii.) Find the longest word from file. iv.) Get the size of the file. v.) Read two lines from file and combine 1st and 2nd line.
86	Class: i.) WAP a small Student Class program. ii.) WAP Constructor(__init__) and Class Variable. iii.) WAP person Account class with Methods like deposit, withdraw, transfer, Variable and Constructor. iv.) WAP Class List CURD. v.) WAP Single Inheritance for class person and student.
87	Database: i.) MySQL - Menu-Driven Database CURD Operations : MENU Options: 1. Show Databases 2. Create Database 3. Create table 4. Insert Data 5. Update Data 6. Delete Data 7. Show/Display Data 8. Exit

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Tkinter:

i.) Tkinter GUI CURD App

1. Create Database [if not exist]
2. Create table [if not exist]
3. Insert Data
4. Update Data
5. Delete Data
6. Show/Display Data

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i.) WAP to count string length without using 'len' function.

ii.) WAP to print all possible combination from the three digits.

iii.) WAP to check the given number is leap year.

iv.) WAP to check whether the given integer is a multiple of both 5 and 7.

v.) WAP to display all integers within the range 100-200 whose sum of digit is an even number.

vi.) WAP to find out the number of vowels in file.

vii.) WAP to reverse the content of file.

viii.) WAP to append data and read data from a file.

ix.) WAP to check whether file exist or not and if exist then read data.

x.) WAP to read a group of string in a text file.

xi.) WAP to write a group of string in a text file.

xii.) WAP to create a menu driven program that:

- A) Add records of participants
- B) Display record
- C) Count participants
- D) Count participants on the basis of event
- E) Count participants on the basis of gender

xiii.) WAP to print the limited data from a file

xiv.) WAP that create an object in a class.

xv.) Create a DataFrame from CSV file using pandas

xvi.) Plot a bar chart using matplotlib